

SHARP NOISE THRESHOLDS AND WIDTH LAWS FOR STOCHASTIC REALIZATIONS OF COMPACT GROUP EXPERIMENTS

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ABSTRACT. We classify bounded-width stochastic realizations of finite continuous numerical interfaces on compact groups with finitely many connected components. The output may be a categorical law or, more generally, a point of a compact convex set in a finite-dimensional normed space. The exact error threshold is the largest constrained Chebyshev radius of a component image. A stationary permutation realization attains the boundary; below it every fixed width has a horizon-independent occupation bound, even when all other registers are unrestricted. The theorem allows both clock-dependent rows and redesign at every horizon, and requires executable returns of all sufficiently large lengths rather than an identity letter. Finite representative functions and positive localization give an intrinsic proof without a faithful matrix embedding. For a planar alphabet of s rotations whose angle vector is dual badly approximable and whose signal amplitude is strictly below one, the minimum width is $\Theta(N^{s/(2s+1)})$ at every fixed error below the sharp threshold; an exact polygon realization supplies the matching upper bound. For rational orthogonal commands and polynomial categorical interfaces, the threshold, a boundary machine, and subcritical occupation bounds are computable. The general continuous classification is an existence theorem, not an algorithm for unencoded real data.

1. THE RADIUS THRESHOLD

A finite stochastic register can carry infinitely many probability distributions. Thus an infinite numerical orbit is not, by itself, an obstruction to finite positive realization. A further difficulty arises when the realization may be redesigned at every horizon and its transitions may depend on a free external clock. The obstruction established here survives both freedoms. It is measured by the radius of an observable image inside its allowed output set.

Let H be a compact Hausdorff group with finitely many connected components. Put $K = H^\circ$ and $C = H/K$. Then K is an open normal subgroup and C is a finite group. Fix a finite alphabet $\mathcal{A}$ with elements $u_a \in H$ whose generated group is dense in H . Our convention is

$$u_{a_1 \cdots a_n} = u_{a_n} \cdots u_{a_1}.$$

The identity e need not be an alphabet element. Instead assume the following executable return condition:

(1.1)

there is an integer $g \geq 0$ such that, for every $n \geq g$, some word of length n has product e .

An identity letter gives $g = 0$. The condition concerns physical products; the stochastic rows on a return word may perform arbitrary processing.

For nonempty finite seed and query sets $\mathcal{S}, \mathcal{J}$, let

$$F_{s,j} : H \longrightarrow Y_j$$

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be continuous, where Y_j is a nonempty compact convex subset of a finite-dimensional real normed space $(V_j, \|\cdot\|_j)$. A query is selected only after the command word. A prescribed joint law of finitely many outputs is treated as one query, not as a collection of marginal constraints. In particular, a categorical interface has

$$Y_j = \Delta_{M_j} = \{p \in \mathbb{R}^{M_j} : p_i \geq 0, \sum_i p_i = 1\}, \quad \|x\|_j = \frac{1}{2} \sum_i |x_i|.$$

Then $\|p - q\|_j$ is total variation. Binary means correspond to $Y = [-1, 1]$ with $\|x\| = |x|/2$.

Definition 1.1 (Clocked realization). At horizon N choose finite nonempty registers S_t , $0 \leq t \leq N$, initialization rows E_s , row-stochastic command matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal decoder maps $D_j : S_N \rightarrow Y_j$. The realized output vector on $w = a_1 \cdots a_N$ is

$$E_s T_{1,a_1} \cdots T_{N,a_N} D_j.$$

Its error is the maximum, over all seeds, length- N words, and queries, of its distance from $F_{s,j}(u_w)$. The command width is $\max_t |S_t|$. Write $W_{N,\varepsilon}$ for its least possible value at error at most ε .

All retained private information, including randomness kept for future use, is charged to the current label. The rows may depend on t and N , but not on an unrecorded input history or on the future query. The external clock and horizon, arbitrary real table descriptions, their construction and lookup, real arithmetic, and exact sampling are uncharged. This is a numerical realization invariant, not uniform workspace or program size. For a finite categorical interface, charging the terminal answer register changes the peak to $\max(W_{N,\varepsilon}, \max_j M_j)$. Thus all boundedness and growth results are unchanged; the binary convention in the preceding manuscript charges at least two labels.

For $A \subset Y_j$ define its constrained radius by

$$\text{rad}_{Y_j}(A) = \min_{y \in Y_j} \sup_{x \in A} \|x - y\|_j.$$

The minimum is attained when A is compact. The critical error is

$$(1.2) \quad \varepsilon_c = \max_{s,j,c \in C} \text{rad}_{Y_j}(F_{s,j}(c)).$$

Here a component is identified with its coset in H . The requirement that the center belong to Y_j is essential for categorical outputs.

Theorem 1.2 (Intrinsic radius classification). *Under (1.1), for every $\varepsilon \geq 0$ the following are equivalent:*

- (1) $\sup_N W_{N,\varepsilon} < \infty$;
- (2) *one stationary finite-state realization with permutation command updates has error at most ε at every horizon;*
- (3) $\varepsilon \geq \varepsilon_c$.

At the boundary, at most $|\mathcal{S}||C|$ command labels suffice. If $\varepsilon < \varepsilon_c$, then for every integer $k \geq 1$ there is an integer $L_k < \infty$ such that every error- ε realization satisfies

$$(1.3) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_k.$$

All intervening widths are unrestricted. In particular, $W_{N,\varepsilon} \rightarrow \infty$.

Neither a positive lower bound on state masses nor a reachable-rank, observability, or conditioning assumption occurs in the theorem. The upper realization is deterministic until its terminal decoder. It need not have the minimum width among all stochastic realizations.

For binary means, (1.2) reduces to one quarter of the largest componentwise oscillation. For a genuine categorical law it is generally not half its diameter; an analytic three-outcome example below has critical total-variation error $2/3$, whereas every individual event has binary threshold

1/2. The proof uses finitely many localized vector tests rather than a separate midpoint for each coordinate.

The finite-component assumption does not assert that H is a Lie group. For example, take $H = \prod_{n \geq 1} \mathbb{R}/\mathbb{Z}$, choose an element whose every finite coordinate tuple together with 1 is rationally independent, and include this element and e as commands. The positive orbit is dense and H is connected but has no faithful finite-dimensional representation. The theorem applies to every finite continuous interface on this group. With infinitely many components the component construction need not be finite; that different case is not included here.

There is also a quantitative counterpart. For planar rotations with a dual badly approximable s -tuple of angles and signal amplitude $0 < \rho < 1$, Theorem 4.1 proves

$$W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}) \quad \text{for every fixed } 0 \leq \varepsilon < \rho/2.$$

The upper realization is exact even when $\varepsilon > 0$. The new lower estimate reaches the entire subcritical interval, by combining harmonic localization with the arithmetic orbit-packing mechanism. This quantitative theorem and the radius classification have distinct hypotheses: algebraic matrix entries alone do not imply the badly approximable angle condition.

1.1. Executable tests. Let $\mathcal{P} = \{u_w : w \in \mathcal{A}^*\}$. We use standard Haar averaging and the density of matrix coefficients of finite-dimensional unitary representations, in realified form [8]. No quantitative mixing assumption for a preassigned random walk is imposed.

Lemma 1.3 (Positive words and common lengths). *The semigroup $\mathcal{P}$ is dense in H , and $\mathcal{P} \cap K$ is dense in K . Every finite subset of $\mathcal{P} \cap K$ has executable representatives of a common positive length.*

Proof. For an element u of a compact group, positive powers return arbitrarily close to e with exponent at least two. Indeed cover the compact closure of its powers by finitely many translates of a sufficiently small neighborhood. Among sufficiently many powers, three lie in one member of the cover; the first and last exponents differ by at least two, and their quotient is arbitrarily close to e . This argument also covers finite-order elements by taking a sufficiently large multiple of the order. It follows that u^{-1} is in the closure of positive powers. The closed semigroup $\overline{\mathcal{P}}$ therefore contains the group generated by the letters, and hence equals H . Since K is open, a relatively open subset of K is open in H ; density gives the second assertion. For finitely many representative lengths ℓ_i , choose $B \geq \max_i \ell_i + g$, $B \geq 1$, and append return words of lengths $B - \ell_i$. The products are unchanged. $\square$

Lemma 1.4 (Finite executable gap). *Let $R : K \rightarrow O(E)$ be a continuous representation on a nonzero finite-dimensional Euclidean space, with no nonzero fixed vector. For every $k \geq 1$ there are $B \geq 1$, a probability law ν on finitely many executable length- B words with products in K , and $d \in (0, 1)$ such that*

$$(1.4) \quad \mathbb{E}_{w \sim \nu} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - d$$

for all unit vectors $u, v_1, \dots, v_k \in E$.

Proof. On the compact product of unit spheres consider the minimum

$$\gamma = \min_{u, v_1, \dots, v_k} \int_K \left(1 - \max_i \langle v_i, R(h)u \rangle \right) d\mu(h).$$

Repeated centers are permitted. The integrand is nonnegative. If a minimizing tuple had integral zero, full support of Haar measure would give $R(h)u \in \{v_1, \dots, v_k\}$ for every h . The orbit of u is connected, hence a singleton, contradicting the fixed-vector hypothesis. Thus $\gamma > 0$.

Partition the compact image $R(K)$ into finitely many measurable sets of small diameter in the Euclidean operator norm, and replace their Haar masses by atoms at nearby executable products

in K . Cells of zero Haar mass can be discarded. The function $A \mapsto \max_i \langle v_i, Au \rangle$ is uniformly 1-Lipschitz in that norm. An approximation radius less than $\gamma/2$ therefore preserves a positive defect uniformly in the unit vectors. Lemma 1.3 gives a common length. Decreasing the defect to a number in $(0, 1)$ proves the assertion. $\square$

Lemma 1.5 (Conditional-centroid contraction). *Let $Z \in E$ be a bounded random physical vector and S the current register. Apply an independent word $w \sim \nu$ from Lemma 1.4, and let the actual block kernel generate a terminal state S' having at most k labels. Then*

$$(1.5) \quad \mathbb{E} \|\mathbb{E}[R(u_w)Z \mid S']\| \leq (1 - d) \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

A fixed intervening return word cannot increase the same conditional norm, regardless of its intermediate or terminal widths.

Proof. Put $z_s = \mathbb{E}[Z \mid S = s]$ and write $T_w(s, i)$ for the actual composite row of the block. Since this row depends on the past only through S ,

$$\Pr(S' = i) \mathbb{E}[R(u_w)Z \mid S' = i] = \sum_s \Pr(S = s) \mathbb{E}_w T_w(s, i) R(u_w) z_s.$$

Choose v_i in the direction of each nonzero terminal centroid, arbitrarily otherwise. Pair, sum in i , and use stochasticity to bound the sum by

$$\sum_s \Pr(S = s) \mathbb{E}_w \max_i \langle v_i, R(u_w) z_s \rangle \leq (1 - d) \sum_s \Pr(S = s) \|z_s\|.$$

Zero-probability states contribute zero. No restriction was placed on registers inside the block. On a fixed return word the physical vector is unchanged at its endpoints; the new state is a stochastic image of the old one, so conditional Jensen gives nonincrease. Independence is used only at complete block boundaries. $\square$

2. REPRESENTATIVE FUNCTIONS AND LOCALIZED RADII

A real representative function on K is a matrix coefficient of a finite-dimensional continuous orthogonal representation, or a finite sum of such coefficients. Their algebra contains the constants, is closed under multiplication by tensor products, and is uniformly dense in $C(K, \mathbb{R})$ by the Peter–Weyl approximation theorem [8]. A finite collection of representative functions can be written on one space

$$\mathcal{V} = \bigoplus_{i=1}^a \text{End}(E_i), \quad \Phi(h) = (R_i(h))_{i=1}^a,$$

where the scalar product is the sum of Frobenius products. Left multiplication defines an orthogonal representation R on $\mathcal{V}$ and satisfies $\Phi(gh) = R(g)\Phi(h)$. Adjoin a trivial summand when necessary. Let

$$P = \int_K R(h) d\mu(h), \quad E = (I - P)\mathcal{V}, \quad Z(h) = (I - P)\Phi(h), \quad Q_0 = \|Z(e)\|.$$

Haar averaging is the orthogonal projection onto fixed vectors. Consequently $Z(gh) = R(g)Z(h)$, the representation on E has no fixed vectors, and $P\Phi(h) = P\Phi(e)$ for $h \in K$.

Choose and fix a coefficient vector a_p for each scalar representative function p , and a linear coefficient map A_G for each vector-valued representative function G . Their norms need not be intrinsic; fixing any such choices suffices. For an arbitrary joint distribution of $h \in K$ and a finite state S , put $Q = \mathbb{E} \|\mathbb{E}[Z(h) \mid S]\|$ and $\bar{p} = \int p d\mu$. The fixed/moving decomposition gives

$$(2.1) \quad \mathbb{E}[p(h) \mid S] = \bar{p} + \langle a_p, \mathbb{E}[Z(h) \mid S] \rangle,$$

$$(2.2) \quad \|\mathbb{E}p - \bar{p}\| \leq \|a_p\|Q, \quad \|\mathbb{E}G - \bar{G}\| \leq \|A_G\|Q.$$

Here the norm of A_G is the operator norm from the Euclidean feature space to the indicated output space. If $D : S \rightarrow Y$ and $M_Y = \max_{y \in Y} \|y\|$, then

$$(2.3) \quad \|\mathbb{E}[p(h)D(S)] - \bar{p}\mathbb{E}D(S)\| \leq M_Y \|a_p\| Q.$$

Indeed condition on S in (2.1), multiply by $D(S)$, and apply the triangle inequality. These are moment estimates, not total-variation convergence of a hidden physical distribution.

Lemma 2.1 (Finite localization of a radius). *Let $f : K \rightarrow Y$ be continuous, with Y compact and convex in a finite-dimensional normed space. Suppose $\text{rad}_Y(f(K)) > \varepsilon$. There are a vector-valued representative function G , a number $\eta > 0$, and nonnegative representative functions $p_1, \dots, p_a$ of Haar mean one such that*

$$\|G - f\|_\infty < \eta, \quad r_* := \min_{y \in Y} \max_i \|p_i \bar{G} - y\| > \delta := \varepsilon + \eta.$$

Proof. Choose $\varepsilon < t < \text{rad}_Y(f(K))$. For each $y \in Y$ some $h \in K$ has $\|f(h) - y\| > t$. The corresponding open subsets of Y cover Y , so a finite subcover yields $h_1, \dots, h_a$ with

$$\min_{y \in Y} \max_i \|f(h_i) - y\| > t.$$

The strict inequality follows also at a minimizing y , since the maximum is everywhere greater than t and Y is compact.

Fix a sufficiently small positive η . Approximate the coordinates of f uniformly by representative functions to obtain $\|G - f\|_\infty < \eta$. For each i , choose a nonzero nonnegative continuous bump supported in a neighborhood where f differs from $f(h_i)$ by less than η . Such a bump exists by normality of the compact Hausdorff space; it has positive Haar integral by full support. Approximate its square root uniformly by a representative function, square the approximation, and normalize its Haar integral. The resulting nonnegative representative function p_i has mean one, and its weighted f -mean can be made arbitrarily close to that of the bump. Thus we may ensure

$$\|p_i \bar{G} - f(h_i)\| < 3\eta.$$

The constrained radius changes by at most the maximum perturbation of the finitely many points. Taking $4\eta < t - \varepsilon$ gives $r_* > \varepsilon + \eta$. $\square$

Proposition 2.2 (Residual moment from a vector interface). *Fix the data of Lemma 2.1. Choose one feature space representing all p_i and $p_i G$, and fix their coefficient choices before defining*

$$L = \max_i (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|), \quad \Delta = r_* - \delta > 0.$$

Suppose a terminal decoder $D(S) \in Y$ obeys

$$\|\mathbb{E}[D(S) \mid w] - G(h(w))\| \leq \delta$$

for every complete deterministic word in a test law. Then $L > 0$, $E \neq \{0\}$, $Q_0 > 0$, and

$$(2.4) \quad Q \geq \Delta/L.$$

Proof. For $p = p_i \geq 0$, wordwise correctness and the triangle inequality imply

$$\|\mathbb{E}[p(h)D(S)] - \mathbb{E}[p(h)G(h)]\| \leq \delta \mathbb{E}p(h).$$

This is a conditional mean error, not a pointwise bound on the random decoder. Write $y = \mathbb{E}D(S)$, which belongs to Y by convexity. Equations (2.2)–(2.3), together with $\bar{p} = 1$, give

$$\|y - p_i \bar{G}\| \leq \delta + (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|)Q.$$

Maximizing in i yields $r_* \leq \delta + LQ$. Hence $LQ \geq \Delta > 0$. In particular the moving feature and its initial norm cannot vanish: by equivariance, $Q_0 = 0$ would imply $Z(h) = 0$ for all h and therefore $Q = 0$. $\square$

2.1. Proof of Theorem 1.2. Choose (s, j, c) with component radius greater than ε . By density there is an executable prefix v with $u_v \in c$. Put $b = |v|$ and $f(h) = F_{s,j}(hu_v)$ on K . Since $Ku_v = c$, this map has the same constrained radius. Apply Lemma 2.1 and Proposition 2.2.

The moving feature is nonzero independently of any particular test law. Otherwise all represented functions would be constant on K ; since each p_i has mean one, this would force $p_i = 1$ and G constant. Then all $p_i G$ would equal a point at distance less than η from Y , contradicting $r_* > \delta \geq \eta$.

For a fixed width k , apply Lemma 1.4 to the resulting moving feature. Let B, d be its block parameters and set

$$\chi = -\log(1 - d) > 0, \quad A = \max(1, LQ_0/\Delta).$$

Start a test word with v ; between independent length- B test blocks use fixed return words. The accumulated physical product is hu_v . Initially $h = e$, so the conditional feature norm after the prefix equals Q_0 , regardless of the hidden-state distribution. If J block endpoints have width at most k , Lemma 1.5 gives

$$Q_N \leq Q_0(1 - d)^J.$$

The machine is correct on every deterministic word of length N . Thus any external law supported on such words inherits the conditional mean bound, with error $\delta = \varepsilon + \eta$ for G . Proposition 2.2 implies

$$(2.5) \quad J\chi \leq \log A.$$

The law depends on the advertised width profile, not on the machine's private state. When $A = 1$, (2.5) excludes even one selected narrow block; no logarithmic loss has occurred.

For clarity we give the spacing count with the full return hypothesis. Discard narrow cuts before $b + B + g$ and after $N - g$. From the remaining cuts greedily choose the first and then the next at distance at least $B + g$. Put an independent block immediately before each selected cut. The initial gap, all intervening gaps, and the final suffix have lengths at least g , so (1.1) fills them with executable returns. At most $b + B + 2g - 1$ cuts were discarded. Each chosen endpoint accounts for at most $B + g$ eligible cuts. Therefore one may take

$$(2.6) \quad L_k = \left\lceil b + B + 2g - 1 + (B + g) \frac{\log A}{\chi} \right\rceil.$$

When the eligible interval is empty the same estimate follows directly by counting. This proves occupation without restricting any register inside a block or gap. If the entire peak is at most k , then $N \leq L_k$, which proves divergence directly. Monotonicity in the horizon is not needed when an identity letter is absent.

For the upper bound, use labels $(s, c) \in \mathcal{S} \times C$, including any labels not reached at a given horizon. Initialize at (s, K) and update by left multiplication

$$(s, c) \mapsto (s, [u_a]c).$$

Choose a minimizing center of $F_{s,j}(c)$ in Y_j as decoder at (s, c) for query j . These are legal decoder values, and the error is at most (1.2). All command updates are permutations of one finite state set, independent of clock and horizon. This proves boundary attainment and the equivalence. $\square$

Proposition 2.3 (A finite return certificate). *Condition (1.1) follows if there are finitely many executable identity-product words whose positive lengths have greatest common divisor one.*

Proof. Let the lengths be $\ell_1, \dots, \ell_a$. Their nonnegative sums modulo ℓ_1 form the subgroup generated by their residues: a subsemigroup of a finite group is a subgroup. The gcd hypothesis makes this the whole residue group. Choose one nonnegative sum n_r for each residue r modulo ℓ_1 , and put $g = \max_r n_r$. Every $n \geq g$ equals its representative n_r plus a nonnegative multiple of ℓ_1 . Concatenating the corresponding return words proves the claim. $\square$

For example, the alphabet $\{R_\theta, R_{-\theta}, R_{2\pi/3}\}$, with $\theta/2\pi$ irrational, contains no identity letter but has return lengths two and three. It satisfies (1.1) with $g = 2$. A single irrational rotation letter does not: there is only one word at each horizon, and a horizon-specific constant decoder can store its numerical answer with one command label. Thus synchronization cannot be silently omitted. Nor is a bounded irreversible semigroup enough: for the identity command $I = 1$ and contraction command $a = \lambda \in (0, 1)$, the mean $\rho\lambda^{\#a(w)}$ has an exact alive/absorbed two-state realization. The group generated with inverses is not bounded. The present theorem is not a compact-semigroup classification.

3. BINARY, CATEGORICAL, AND OBSERVABLE INTERFACES

Corollary 3.1 (Component oscillation for binary means). *For a finite continuous binary-mean interface $f_{s,j} : H \rightarrow [-1, 1]$, with the return hypothesis of Theorem 1.2, the exact bounded-width threshold in total variation is*

$$(3.1) \quad \varepsilon_c = \frac{1}{4} \max_{s,j,c} \left(\max_{h \in c} f_{s,j}(h) - \min_{h \in c} f_{s,j}(h) \right).$$

It is attained by a stationary permutation realization. Below it every fixed width has the occupation bound (1.3).

Proof. In the norm $|\cdot|/2$, the radius of an interval is one quarter of its length. Its midpoint lies in $[-1, 1]$. Apply Theorem 1.2. $\square$

This recovers the scalar component theorem, including all of its nonuniform quantifiers. The vector theorem does not follow by applying this corollary separately to coordinates, because the coordinate centers need not belong to the output simplex.

Proposition 3.2 (A three-outcome threshold). *On the circle let $\omega = e^{2\pi i/3}$ and, for $j = 0, 1, 2$, set*

$$p_j(\phi) = \frac{1}{9} \left| 1 + e^{i\phi}\omega^{-j} + e^{2i\phi}\omega^{-2j} \right|^2.$$

For any dense circle command alphabet satisfying the return condition, the categorical law $p = (p_0, p_1, p_2)$ has critical total-variation error $2/3$. Each separately requested binary event $\{j\}$ has threshold $1/2$.

Proof. Orthogonality of the three characters gives $\sum_j p_j = 1$; each entry is nonnegative. At $\phi = 2\pi j/3$ the vector is the simplex vertex e_j . Hence every center $q \in \Delta_3$ has maximum distance at least

$$\max_j \text{TV}(e_j, q) = 1 - \min_j q_j \geq 2/3.$$

The uniform center has distance at most $2/3$ from every point of Δ_3 , by convexity and the values at its vertices. Thus the constrained radius is exactly $2/3$. On the other hand each p_j has range $[0, 1]$, so its binary mean $2p_j - 1$ has oscillation two. Apply Theorem 1.2 and Corollary 3.1. In particular the categorical radius exceeds half the diameter, which here equals $1/2$. $\square$

Corollary 3.3 (Joint measure-once unitary laws). *Let unitary commands have compact closure H and satisfy the return condition. For a fixed unit seed ψ_s and a finite measurement $E_{j,l} \geq 0$, $\sum_l E_{j,l} = I$, prescribe the law*

$$F_{s,j}(h)_l = \langle h\psi_s, E_{j,l}h\psi_s \rangle.$$

The critical error is its componentwise simplex radius in total variation. For closure $U(d)$ and one orthonormal-basis measurement, this radius is $1 - 1/d$.

Proof. The displayed map is a continuous simplex-valued polynomial in realified matrix entries. A closed unitary matrix group has finitely many components, so Theorem 1.2 applies. For $U(d)$, every unit vector is reachable from the seed, and its squared coordinate magnitudes range over the whole simplex. The simplex vertices force radius at least $1 - 1/d$, attained by the uniform distribution. $\square$

A query selects one complete measurement, not simultaneous answers to incompatible measurements. Several compatible finite outputs, with specified correlations, instead form one joint categorical law. Language recognition at a cutpoint is a different requirement.

For linear binary interfaces on a compact matrix group, write

$$f_{s,j}(h) = q_j^T h x_s \in [-1, 1], \quad V_r = \text{span}\{h x_s : h \in H, s \in \mathcal{S}\},$$

$$V_n = \{x \in V_r : q_j^T h x = 0 \text{ for all } h \in H, j \in \mathcal{J}\}.$$

Both spaces are invariant. The observable space $V_o = V_r \cap V_n^\perp$ is the orthogonal complement of V_n inside V_r , canonically realizing the observable quotient.

Theorem 3.4 (Intrinsic zero-error criterion). *For these linear means, bounded exact clocked width is equivalent to finiteness of the generated group's image in $O(V_o)$, and to trivial action of K on V_o . An exact stationary permutation realization then exists.*

Proof. Corollary 3.1 identifies bounded exact width with constancy of all interface means on every component. Under that condition, for $k \in K$ and $g, h \in H$, normality gives

$$q_j^T h(k - I)g x_s = q_j^T (h k h^{-1})h g x_s - q_j^T h g x_s = 0.$$

Thus $(k - I)V_r \subset V_n$. On the invariant orthogonal complement V_o the difference also lies in V_o , hence vanishes. The observable action factors through the finite component group. Conversely, if K acts trivially on V_o , the same difference is invisible and all means are componentwise constant. A finite image of the dense generated group has the same finite image closure from H , whose connected subgroup K acts trivially. The zero-dimensional quotient is included. $\square$

Corollary 3.5 (The planar threshold). *Let an orthogonal planar alphabet contain I and an irrational rotation, with any additional orthogonal letters. For signed coordinate seeds $\pm p e_1, \pm p e_2$, coordinate queries, and $0 < \rho \leq 1$, the threshold is $\varepsilon_c = \rho/2$. At and above it one command label with a fair binary decoder suffices. Below it every fixed width has a finite occupation budget.*

Proof. The identity component is $SO(2)$, and every specified mean has range $[-\rho, \rho]$ on each component of $H = SO(2)$ or $O(2)$. All midpoint decoders equal zero, so the seed/component register collapses to one label. $\square$

For the noncommuting rational alphabet

$$I, \quad R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

one has $FRF = R^{-1} \neq R$. The rotation has infinite order: otherwise $(3 + 4i)/5$ and its inverse would be algebraic integers, whereas their rational sum $6/5$ is not an integer. This example has the sharp threshold above. We do not infer a badly approximable angle from rational matrix entries.

Proposition 3.6 (Independent toral coordinates). *Suppose a component is identified with the full product torus $(\mathbb{R}/2\pi\mathbb{Z})^m$, surjectively in independent coordinates, and a mean on it equals*

$$a_0 + \sum_{l=1}^m (a_l \cos \theta_l + b_l \sin \theta_l).$$

Its contribution to (3.1) is $\frac{1}{2} \sum_l \sqrt{a_l^2 + b_l^2}$.

Proof. Each summand has the symmetric range with amplitude $\sqrt{a_l^2 + b_l^2}$, and all extrema are simultaneously attainable in the full product. Divide the resulting oscillation by four. $\square$

On a proper subtorus, extrema must instead be taken subject to its relations. Ambient coordinates do not certify independence. Similarly, invisible directions do not create an obstruction: for $H = \{\text{diag}(R_\theta, 1)\}$, seed and query e_3 give constant mean one, whereas seed ρe_1 and query e_1 have threshold $\rho/2$.

Proposition 3.7 (Compact similarity). *The classification applies to a finite alphabet in $GL(D, \mathbb{R})$ whose generated group, including inverses, is uniformly bounded, with the executable return hypothesis retained. Conjugating to an invariant Euclidean metric changes neither output probabilities nor stochastic widths.*

Proof. Uniform bounds on matrices and inverses make the closure a compact group. The matrix $Q = \int_H h^T h d\mu(h)$ is positive definite and satisfies $A^T Q A = Q$. Hence $Q^{1/2} A Q^{-1/2}$ is orthogonal. Transport the continuous output maps through this conjugacy; for a linear mean transport x_s to $Q^{1/2} x_s$ and q_j to $Q^{-1/2} q_j$. Exact identity products and their word lengths are preserved. $\square$

4. MATCHING WIDTH LAWS THROUGHOUT THE SUBCRITICAL INTERVAL

We now distinguish the dimension of the command arithmetic from the dimension of the physical orbit. Let $s \geq 1$ and let the alphabet contain the rotations $R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_s}$ and I . For the matching upper bound the alphabet is exactly these letters, optionally with the reflection $F = \text{diag}(1, -1)$. Use signed coordinate seeds of radius ρ and coordinate queries. Extra letters do not affect the lower bounds, but are not silently included in the upper construction.

For a real number x , write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The dual Diophantine condition is

$$(4.1) \quad \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|n\|_1^{-\tau} \quad (0 \neq n \in \mathbb{Z}^s), \quad b > 0, \quad \tau \geq s.$$

In particular $1, \alpha_1, \dots, \alpha_s$ are rationally independent. When $\tau = s$ this is the dual badly approximable condition. All implicit constants below may depend on $\alpha, b, s, \rho, \varepsilon$, but not on N or a stochastic realization.

Theorem 4.1 (Full-subcritical matching law). *Suppose (4.1) holds with $\tau = s$, and let $0 < \rho < 1$. For every fixed $0 \leq \varepsilon < \rho/2$,*

$$(4.2) \quad W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}).$$

The upper bound is achieved by an exact horizon-dependent realization. At $\varepsilon \geq \rho/2$, one command label suffices. If $\tau \geq s$ is arbitrary, the lower bound $W_{N,\varepsilon} = \Omega(N^{s/(2\tau+1)})$ still holds throughout the same subcritical interval.

The exponent for one badly approximable rotation is $1/3$, not logarithmic. The statement includes zero error when the signal has strict amplitude slack $\rho < 1$. The threshold theorem continues to include $\rho = 1$, but the exact polynomial upper bound here is not asserted at that endpoint. The earlier arithmetic argument obtained this scale in a smaller noise range. The proof below supplies the missing full-noise lower estimate and states the exact upper realization in full.

Proposition 4.2 (An explicit family in every command dimension). *Let ξ be a real algebraic integer of degree $s + 1$. Then $\alpha = (\xi, \xi^2, \dots, \xi^s)$ satisfies (4.1) with $\tau = s$ and some effectively computable $b > 0$.*

Proof. For $0 \neq n \in \mathbb{Z}^s$ put $H = \|n\|_1$, and choose $n_0 \in \mathbb{Z}$ so that $z = n_0 + \sum_{l=1}^s n_l \xi^l$ has modulus $\|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}$. The degree assumption gives $z \neq 0$. Write $\xi_1 = \xi, \dots, \xi_{s+1}$ for the conjugates and put $M_j = \max_{1 \leq l \leq s} |\xi_j^l|$. Then $|n_0| \leq HM_1 + 1/2$ and

$$\left| n_0 + \sum_{l=1}^s n_l \xi_j^l \right| \leq H(M_1 + M_j + 1/2) \quad (j \geq 2).$$

The norm of the nonzero algebraic integer z is a nonzero integer. Thus

$$|z| \geq H^{-s} \prod_{j=2}^{s+1} (M_1 + M_j + 1/2)^{-1}.$$

Choose a positive rational b below this computable positive constant and $1/2$. This proves the claim. It concerns algebraic *angles*, not algebraic entries of their rotation matrices. $\square$

4.1. Harmonic localization. For the harmonic calculation assume that the alphabet contains I, R_θ and that one seed-query pair has mean $\rho \cos \phi$ after accumulated angle ϕ , where $0 < \rho \leq 1$. Other seeds, queries, or commands impose additional constraints and do not invalidate the lower bound. Let $\lambda = e^{i\theta}$ have infinite order. Put $\delta = 2\varepsilon < \rho$, and choose an integer

$$(4.3) \quad r \geq 1, \quad r > \frac{\delta}{\rho - \delta}, \quad m = r + 1.$$

For example $r = \lfloor \delta/(\rho - \delta) \rfloor + 1$ works. Define

$$(4.4) \quad \Delta_r = 2 \left(\frac{\rho r}{r+1} - \delta \right) > 0, \quad A_r = \frac{2(1 + \rho + \delta)(2r+1)m}{\Delta_r} > 1.$$

Lemma 4.3 (Explicit harmonic localization). *The nonnegative trigonometric polynomials*

$$p_{\pm}(\phi) = \frac{(1 \pm \cos \phi)^r}{2^{-r} \binom{2r}{r}}$$

have Haar mean one and weighted means

$$\int p_{\pm}(\phi) \rho \cos \phi \frac{d\phi}{2\pi} = \pm \frac{\rho r}{r+1}.$$

For the moving feature vector

$$Z_m(\phi) = (\cos \phi, \sin \phi, \dots, \cos m\phi, \sin m\phi),$$

any terminal realization with wordwise mean error at most δ obeys

$$(4.5) \quad \mathbb{E} \|\mathbb{E}[Z_m(\phi) \mid S]\| \geq \frac{\Delta_r}{2(1 + \rho + \delta)(2r+1)}.$$

Proof. The identity $1 + \cos \phi = 2 \cos^2(\phi/2)$, or the constant Laurent coefficient, gives

$$C_r := \int (1 + \cos \phi)^r \frac{d\phi}{2\pi} = 2^{-r} \binom{2r}{r}, \quad C_{r+1}/C_r = \frac{2r+1}{r+1}.$$

Hence the weighted cosine mean is $C_{r+1}/C_r - 1 = r/(r+1)$. Translation by π gives the negative case.

We use the unweighted real coefficient vector in the basis $1, \cos \phi, \sin \phi, \dots$, without orthonormal Fourier rescaling. The absolute sum of the real cosine coefficients of either $p_{\pm}$, including the constant coefficient, is

$$B_r = \frac{4^r}{\binom{2r}{r}} \leq 2r + 1.$$

For p_+ its coefficients are nonnegative and their sum is $p_+(0)$; for p_- they differ only by alternating signs. The inequality follows because the central binomial coefficient is the largest of the $2r + 1$ coefficients summing to 4^r . Multiplication by $\rho \cos \phi$ has coefficient absolute sum at most ρB_r , since $\cos u \cos v = (\cos(u + v) + \cos(u - v))/2$. Thus the Euclidean norms of the moving coefficient vectors of $p_\pm$ and $p_\pm \rho \cos \phi$ are at most $2r + 1$ and $\rho(2r + 1)$, respectively. For completeness, let $c = \mathbb{E}D$, $Q = \mathbb{E}\|\mathbb{E}[Z_m | S]\|$. For either weight, the conditional coefficient identity gives $|c - \int p_\pm \rho \cos \phi| \leq \delta + (1 + \rho + \delta)(2r + 1)Q$. Subtracting the two inequalities gives (4.5). Correctness is averaged word by word before using this identity. The starting feature norm is $\sqrt{m} \leq m$; A_r deliberately uses the latter bound to obtain a rational closed form. $\square$

The integer r in (4.3) is the localization order; it is unrelated to the number s of rotation letters. For fixed subcritical error it is finite and independent of N and k .

Theorem 4.4 (A full-noise arithmetic occupation bound). *Assume (4.1) and $0 \leq \varepsilon < \rho/2$. Fix the localization parameters r, m, A_r in (4.3)–(4.4). For an integer $k \geq 1$, put*

$$L = \lceil (2k)^{1/s} \rceil, \quad B = s(L - 1).$$

Every error- ε realization of peak at most k satisfies

$$(4.6) \quad N < B \left(1 + b^{-2}(mB)^{2\tau} \log A_r \right).$$

With no restriction on other widths, the number of cuts of width at most k is at most

$$(4.7) \quad \left\lceil B - 1 + Bb^{-2}(mB)^{2\tau} \log A_r \right\rceil.$$

Proof. Use s consecutive groups of $L - 1$ command positions. In group i , choose J_i uniformly from $\{0, \dots, L - 1\}$, apply J_i copies of the i th rotation, and fill the other positions with identities. This defines an external law on length- B words with $L^s \geq 2k$ equally likely products. On the harmonic feature use

$$R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt}).$$

For two distinct count vectors J, J' and any frequency $1 \leq l \leq m$, condition (4.1) and $\|l(J - J')\|_1 \leq mB$ give

$$|e^{2\pi i l(J - J') \cdot \alpha} - 1| = 2|\sin(\pi l(J - J') \cdot \alpha)| \geq 4b(mB)^{-\tau} =: \sigma.$$

Here $2 \sin(\pi x) \geq 4x$ for $0 \leq x \leq 1/2$. Also $b \leq 1/2$, $m \geq 2$, and $B \geq 1$, so $\sigma \leq 1$.

For a unit vector $u = (u_1, \dots, u_m)$ in the frequency blocks, the squared distance between the two represented orbit points is

$$\sum_{l=1}^m \|u_l\|^2 |e^{2\pi i l(J - J') \cdot \alpha} - 1|^2 \geq \sigma^2.$$

An open ball of radius $\sigma/2$ about any unit center contains at most one point. At least half the L^s points therefore have distance at least $\sigma/2$ from all of any k centers. Since the inner-product defect between unit vectors equals half their squared distance, the finite word law has defect at least

$$d_k = \sigma^2/16 = b^2(mB)^{-2\tau}.$$

Apply conditional-centroid contraction to successive blocks and Lemma 4.3 to the terminal query. The starting norm is $\sqrt{m} \leq m$. Thus

$$\left\lfloor \frac{N}{B} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq b^{-2}(mB)^{2\tau} \log A_r,$$

which proves (4.6). The greedy endpoint count, with identity gaps and no prefix, gives (4.7). Internal registers remain unrestricted. Finally $B < s(2k)^{1/s}$, so (4.6) implies the asserted $\Omega(N^{s/(2\tau+1)})$ lower bound. $\square$

4.2. An exact polygon realization. Write $\eta_\alpha(q) = \max_i \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}$. The following elementary construction explains why a common denominator, rather than the number of numerical orbit points, controls the upper width.

Proposition 4.5 (Exact enclosure upper bound). *Let $0 < \rho < 1$, $a = (1 + \rho)/2$, and $\gamma = \log(a/\rho) > 0$. Choose an integer $q_0 \geq 4$ with $\cos(\pi/q_0) \geq a$. If $q \geq q_0$ and*

$$(4.8) \quad 4\pi^2 N \eta_\alpha(q)/q^2 \leq \gamma,$$

then the indicated rotation alphabet, with or without F , has an exact horizon- N realization with q command labels.

Proof. Let P_q be the regular polygon with vertices $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$, $0 \leq j < q$. For each i choose a nearest integer p_i to $q\alpha_i$ and put

$$t = \pi/q, \quad h_i = 2\pi|\alpha_i - p_i/q| \leq t, \quad \Lambda_i = \frac{\cos(t - h_i)}{\cos t}.$$

The facet inequalities of the regular polygon give $R_{2\pi\alpha_i}P_q \subset \Lambda_i P_q$: the nearest facet normal to a rotated vertex has angular difference $t - h_i$. Further,

$$\log \Lambda_i \leq \log(1 + \tan(t)h_i) \leq \tan(t)h_i \leq 4\pi^2 \eta_\alpha(q)/q^2,$$

since $\cos h + \tan(t) \sin h \leq 1 + \tan(t)h$ and $\tan t \leq 2t$ for $0 \leq t \leq \pi/4$. Let $\Lambda = \max(1, \Lambda_1, \dots, \Lambda_s)$. The identity and reflection preserve P_q , so every command U satisfies $\Lambda^{-1}UP_q \subset P_q$.

For each command and each vertex choose a probability row with barycenter $\Lambda^{-1}Uv_j$ in P_q . These are actual common command rows, not independent prefix/suffix factorizations. At most three successors suffice by planar convexity. Initialize the seed ρu_s , $u_s \in \{\pm e_1, \pm e_2\}$, with a row whose barycenter is au_s . This is possible since $a \leq \cos(\pi/q)$, the polygon's inradius. After a length- N word the mean vertex is

$$a\Lambda^{-N}U_w u_s.$$

By (4.8), $\rho\Lambda^N \leq a$. Hence the terminal columns

$$d_j(v) = \frac{\rho\Lambda^N}{a} e_j^T v$$

belong to $[-1, 1]$ and return exactly $\rho e_j^T U_w u_s$. The rows can be stationary within a horizon; the terminal scaling and chosen denominator may depend on N . No one finite all-horizon exact machine is asserted. $\square$

Lemma 4.6 (Comparable simultaneous denominators). *Assume (4.1). For every integer $Q \geq 2$ there is q with $1 \leq q \leq Q^s$, $\eta_\alpha(q) \leq 1/Q$, and*

$$q \geq cQ^{s/(\tau+1-s)}, \quad c = \left(\frac{b}{s^{\tau+1}} \right)^{s/(\tau+1-s)}.$$

In particular, for $\tau = s$ one has $cQ^s \leq q \leq Q^s$.

Proof. Partition the torus into Q^s half-open cubes and apply the pigeonhole principle to $0, \alpha, \dots, Q^s \alpha$. Their difference gives $q \leq Q^s$ and integers p_i with $|q\alpha_i - p_i| \leq 1/Q$.

Put $M = \lfloor q^{1/s} \rfloor$. The $(M+1)^s > q$ vectors in $\{0, \dots, M\}^s$ cannot all have distinct residues $n \cdot p$ modulo q . Subtract two to obtain a nonzero vector n with $q \mid n \cdot p$ and $H := \|n\|_1 \leq sq^{1/s}$. Therefore

$$bH^{-\tau} \leq \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad q^{(\tau+1-s)/s} \geq bQ/s^{\tau+1}.$$

The displayed bound follows. This argument supplies the needed quantitative transference directly; no unspecified denominator subsequence is being used. $\square$

Proof of Theorem 4.1. The lower estimate follows from Theorem 4.4. For $\tau = s$, take the constants a, γ, q_0, c above and an integer

$$Q \geq \max \left(2, (q_0/c)^{1/s}, \left(\frac{4\pi^2 N}{c^2 \gamma} \right)^{1/(2s+1)} \right).$$

Choosing the least such integer, Lemma 4.6 gives $q \leq Q^s = O(N^{s/(2s+1)})$, $q \geq q_0$, and

$$4\pi^2 N \eta_\alpha(q)/q^2 \leq 4\pi^2 N/(c^2 Q^{2s+1}) \leq \gamma.$$

Proposition 4.5 supplies the exact upper bound. At and above $\rho/2$ use the fair decoder from Corollary 3.5. This proves all claims. $\square$

The command count s is an arithmetic parameter, not a replacement for the dimension of the physical circle. The constants deteriorate as $\varepsilon \uparrow \rho/2$ through the localization degree, and as $\rho \uparrow 1$ through the enclosure slack. Both dependences are explicit in the finite inequalities. The theorem makes no uniform claim at these limiting parameters.

5. EFFECTIVE ALGEBRAIC-CIRCLE BOUNDS

For algebraic unit-circle eigenvalues of infinite order, an explicit separation estimate replaces the Diophantine hypothesis of the preceding section. The parameters r, m, A_r below are those of (4.3)–(4.4); in this section r denotes the localization order, not the number of command rotations. The alphabet contains I, R_θ , and one interface mean is $\rho \cos \phi$.

Lemma 5.1 (Separated powers give a block defect). *Suppose $|\lambda^n - 1| \geq \sigma$ for every integer $1 \leq n \leq 2mk$, where $0 < \sigma \leq 1$. Write $R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt})$ for the representation acting on Z_m . On this representation, the uniform law on the $2k$ words with products R_θ^j , $0 \leq j < 2k$, each padded to length $2k$, has defect at least $\sigma^2/16$ in (1.4).*

Proof. Write a unit vector as $u = (u_1, \dots, u_m)$ in its two-dimensional frequency blocks. For distinct indices $i, j \in \{0, \dots, 2k-1\}$,

$$\|R_m(i\theta)u - R_m(j\theta)u\|^2 = \sum_{l=1}^m \|u_l\|^2 |\lambda^{l(i-j)} - 1|^2 \geq \sigma^2.$$

A ball of radius strictly less than $\sigma/2$ about any of k centers contains at most one of these $2k$ points. Thus at least k points have distance at least $\sigma/2$ from all centers (the same conclusion follows by taking the open balls of radius $\sigma/2$). For unit centers the defect is one half of the squared distance to the closest center. Averaging gives at least $(1/2)(\sigma^2/8) = \sigma^2/16$. $\square$

Theorem 5.2 (Full-noise algebraic-circle bound). *Assume λ is algebraic and not a root of unity. Let its primitive irreducible integer polynomial be*

$$P(z) = az^d + a_{d-1}z^{d-1} + \dots + a_0, \quad a > 0.$$

Set $H_P = \max(a, |a_{d-1}|, \dots, |a_0|)$ and

$$B_P = a(1 + H_P/a)^{d-1} > 1.$$

With r, m, A_r as in (4.3)–(4.4), every error- ε realization of peak at most k satisfies

$$(5.1) \quad N < 2k \left(1 + 16 \cdot 2^{2(d-1)} B_P^{4mk} \log A_r \right).$$

Its number of command cuts $t \in \{1, \dots, N\}$ with width at most k , even if other widths are unrestricted, is at most

$$(5.2) \quad 2k - 1 + 32k \cdot 2^{2(d-1)} B_P^{4mk} \log A_r.$$

Consequently, for fixed algebraic data, ρ , and $\varepsilon < \rho/2$,

$$(5.3) \quad W_{N,\varepsilon} \geq \frac{\log N - O(\log \log N)}{4m \log B_P}.$$

All constants in the finite inequalities are explicit. If $\lambda = (a_1 + ib_1)/c$ with integers $a_1, b_1, c > 0$, $a_1^2 + b_1^2 = c^2$, and λ not a root of unity, then (5.1)–(5.3) hold with the factor $2^{2(d-1)}$ deleted and B_P replaced by c .

Proof. List the conjugates with $\lambda_1 = \lambda$. They are distinct by irreducibility in characteristic zero. None is a root of unity: otherwise irreducibility would make P a cyclotomic polynomial up to its primitive sign. The resultant

$$\text{Res}(P, z^n - 1) = a^n \prod_{j=1}^d (\lambda_j^n - 1)$$

is therefore a nonzero integer. Each conjugate has modulus at most $R_P = 1 + H_P/a$. For completeness, if $|z| > R_P$, then

$$\sum_{j=0}^{d-1} |a_j| |z|^j < \frac{H_P |z|^d}{|z| - 1} < a |z|^d,$$

so $P(z) \neq 0$. Using $|z^n - 1| \leq 2 \max(1, |z|)^n$ for each other conjugate, the resultant bound gives

$$(5.4) \quad |\lambda^n - 1| \geq 2^{-(d-1)} \left(a \prod_{j=2}^d \max(1, |\lambda_j|) \right)^{-n} \geq 2^{-(d-1)} B_P^{-n}.$$

For $1 \leq n \leq 2mk$, take $\sigma = 2^{-(d-1)} B_P^{-2mk}$ in Lemma 5.1. The block defect is at least

$$d_k = \frac{2^{-2(d-1)} B_P^{-4mk}}{16}.$$

Repeated conditional contraction and Lemma 4.3 imply

$$\left\lfloor \frac{N}{2k} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq 16 2^{2(d-1)} B_P^{4mk} \log A_r.$$

This proves (5.1). The greedy occupation argument in the proof of Theorem 1.2, with $b = 0$ and block length $2k$, proves (5.2). Taking logarithms gives (5.3): fix $C > 1/(4m \log B_P)$. If $k \geq C \log N$ the claim is immediate; otherwise $\log k \leq \log C + \log \log N$, which can be absorbed in the logarithm of (5.1). The implicit constant may depend on P, ρ, ε , not on N .

In the rational-circle case, $(a_1 + ib_1)^n - c^n$ is a nonzero Gaussian integer, of modulus at least one. Thus $|\lambda^n - 1| \geq c^{-n}$ directly. Repeat the proof with $\sigma = c^{-2mk}$. Here $c > 1$ automatically. A nonminimal supplied denominator is still valid but weakens the bound. $\square$

The threshold is sharp; these width rates are not asserted sharp. The factor m records the increased localization degree as the error approaches the boundary. At smaller errors, the retained coordinate-calibration estimates can have better constants. One should use the larger of applicable lower bounds, rather than claiming numerical improvement at every parameter value.

Proposition 5.3 (An explicit nonrational algebraic example). *Let $t = (1 - \sqrt{13})/2$ and*

$$\lambda = \frac{t + i\sqrt{4 - t^2}}{2}.$$

Then $|\lambda| = 1$, λ is not a root of unity, and one can take $d = 4$ and $B_P = 8$ in Theorem 5.2.

Proof. The number t lies strictly between -2 and 2 and solves $t^2 - t - 3 = 0$. Substitution of $t = z + z^{-1}$ gives

$$P(z) = z^4 - z^3 - z^2 - z + 1.$$

Its reduction modulo two is $z^4 + z^3 + z^2 + z + 1$. It has no linear factor over $\mathbb{F}_2$ and is not divisible by the only monic irreducible quadratic $z^2 + z + 1$; hence it is irreducible. Gauss's lemma proves irreducibility over $\mathbb{Q}$. The conjugate value $(1 + \sqrt{13})/2 > 2$ gives a positive real conjugate of λ greater than one. A root of unity cannot have such a conjugate. Finally $a = H_P = 1$ and $d = 4$, so $B_P = 2^3 = 8$. $\square$

For a reproducible rational calibration, take $\rho = 1/10$ and $\varepsilon = 1/25$. Then $\delta = 2/25$, $r = 5$, $m = 6$, $\Delta_r = 1/150$, and $A_r = 23364$. Thus the rational-circle alphabet above obeys

$$N < 2k(1 + 16 \cdot 5^{24k} \log 23364).$$

This is a concrete bound inside the interval between the former small-error obstruction and the sharp threshold $1/20$. The exact coefficient and resultant checks accompanying the article verify the displayed arithmetic, not the universal theorem by finite experimentation.

6. FINITE ENCODINGS AND BOUNDARY QUOTIENTS

The general continuous theorem is not an algorithm from arbitrary real data. We now give an explicit finite input class for which its conclusions are effective. A rational orthogonal input consists of a finite alphabet in $O(D, \mathbb{Q})$ containing I , finite seed and query sets, and rational polynomials $F_{s,j,l}$ in the matrix entries specifying categorical probabilities. The promise $F_{s,j}(H) \subset \Delta_{M_j}$ can itself be checked once the compact closure is computed. Real algebraic output numbers are encoded by polynomial equations and isolating data.

We use two established algorithmic inputs: computation of a finite defining description of the Zariski closure of a finitely generated rational matrix group [6, 11], and real quantifier elimination with effective semialgebraic connected components [1]. The former is a terminating closure algorithm, not merely an enumeration of valid polynomial identities with an unknown stopping point. Compact orthogonal groups are real algebraic, so the real points of the computed closure in $O(D)$ give the Euclidean closure H [4]. The complex algebraic components must not be substituted for the real connected components used here.

Theorem 6.1 (Effective rational classification). *For a rational orthogonal input, the following are computable:*

- (1) *a semialgebraic description of every connected component of H , the finite group C , and the component of each command;*
- (2) *the critical total-variation error ε_c as an exact real algebraic number, and an attaining stationary component-permutation machine with algebraic decoders;*
- (3) *for each algebraic $0 \leq \varepsilon < \varepsilon_c$ and integer $k \geq 1$, an integer occupation constant L_k valid for every error- ε clocked realization;*
- (4) *for every fixed N and algebraic $\varepsilon \geq 0$, the least width $W_{N,\varepsilon}$ and an algebraic realization attaining it.*

These are terminating computability statements. No polynomial running-time bound, practical implementation of general closure/quantifier elimination, or finite-random-bit cost bound is asserted.

Proof. Compute H using the cited closure algorithm, then its real connected components and algebraic sample points. The component containing I is K . Products of sample points and semialgebraic membership tests determine the component multiplication table and the command images. In particular K is now an effectively given compact semialgebraic group.

For a component c and a query, the condition that q be a radius- t center is the real first-order formula

$$(6.1) \quad q \in \Delta_{M_j}, \quad \forall h \in c : \sum_{l=1}^{M_j} |F_{s,j,l}(h) - q_l| \leq 2t.$$

Absolute values have a finite semialgebraic description. Quantifier elimination gives the set of admissible (q, t) . Its least t exists by compactness and is real algebraic; algebraic sampling supplies an attaining q . Maximize over the finite interface and components. This proves (1)–(2), including equality at the critical error.

We detail effectivity of the occupation proof, since computing the threshold alone would not supply it. Choose a component with radius greater than the given ε , and enumerate positive words until one lies in that component. Its rational product gives the prefix used in the proof of Theorem 1.2. The resulting map $f(h) = F_{s,j}(hu_v)$ is polynomial, so it can be used directly in place of G , with no approximation error.

For any finite tensor degree, all required polynomial functions are coefficients of the explicitly given tensor representation. Its fixed subspace is

$$\{z : \forall h \in K, (R(h) - I)z = 0\}.$$

This linear subspace is effectively semialgebraic. Real algebraic sampling, successively outside the span of the vectors already chosen, gives an algebraic basis in finitely many steps. Its Gram matrix gives the orthogonal projection P exactly. Thus every polynomial Haar moment is computable algebraically from $P\Phi(e)$; numerical integration of Haar measure is unnecessary.

Enumerate finite lists of squares of rational polynomials on K , discarding zero Haar integrals and normalizing the others. For a list p_i , compute $b_i = \int p_i f d\mu$ and its constrained simplex radius by quantifier elimination. Rational polynomials are uniformly dense among real polynomial restrictions on a compact matrix group; the localization proof therefore guarantees that some enumerated list has radius strictly greater than ε . Strict algebraic comparison detects that list. Fix a positive rational Δ below the radius difference and a positive rational upper bound L for all coefficient constants in Proposition 2.2. These give the certified residual $Q \geq \Delta/L$.

Next enumerate finite rational probability laws on executable words of a common length whose products lie in K , together with rational $d \in (0, 1)$. The assertion (1.4), quantified over the finitely many unit vectors in the algebraic moving space, is semialgebraic: each maximum of finitely many inner products can be expressed by a finite disjunction. It is therefore decidable. Lemma 1.4 and sufficiently small rational perturbations of its atom weights ensure that this search terminates. Padding by the identity letter makes every finite choice executable at one common length.

The resulting B, d, L, Δ, Q_0 determine (2.6). To avoid exact comparisons involving logarithms, take a rational $A' \geq \max(1, LQ_0/\Delta)$ and use

$$\frac{\log A}{-\log(1-d)} \leq \frac{A'}{d}.$$

An integer above $b + B - 1 + BA'/d$ is a computable occupation bound. This proves (3).

For (4), fix a candidate width k and use k labels at every cut, padding smaller registers with unused labels. Stochastic rows, initialization, and categorical decoder probabilities are finitely many variables in compact simplexes. For each of the finitely many length- N words, the output vector is polynomial in these variables, and the target vector is rational. The total-variation inequalities are semialgebraic. Quantifier elimination therefore decides feasibility and supplies an algebraic point when feasible. A deterministic register storing the seed and entire prefix is a finite exact realization, so increasing k must terminate. This proves the least-width claim for each finite horizon, without claiming a finite-horizon cutoff for all stationary or nonuniform realization questions. $\square$

The distinction between an effective theorem and an implemented solver matters. The accompanying finite checks verify displayed exact examples and proof constants; they do not execute the general group-closure and quantifier-elimination construction. The finite-bit compiler for rational finite groups remains in the complete predecessor supplement with its separate hypotheses and costs. It is not used to reprice arbitrary algebraic decoders as cost-free finite-bit procedures.

6.1. The least component-quotient realization. The upper bound $|\mathcal{S}||C|$ can often be reduced. Here is an exact finite characterization of that reduction, with its scope stated explicitly. Put $X = \mathcal{S} \times C$. A *component-quotient realization* has a deterministic state of the form $\pi(s, c)$, where $\pi : X \rightarrow Q$ is a surjection; its command update is induced by $(s, c) \mapsto (s, [u_a]c)$. It is stationary and has no additional state recording a representative word within a component.

Call a partition $\mathcal{B}$ of X equivariant if

$$(s, c) \sim (s', c') \implies (s, dc) \sim (s', dc') \quad (d \in C).$$

For a block $B \in \mathcal{B}$ and query j , set

$$A_{B,j} = \bigcup_{(s,c) \in B} F_{s,j}(c).$$

Theorem 6.2 (Finite quotient minimum). *The least number of command labels in an error- ε component-quotient realization is*

$$(6.2) \quad \min \left\{ |\mathcal{B}| : \mathcal{B} \text{ equivariant and } \text{rad}_{Y_j}(A_{B,j}) \leq \varepsilon \text{ for every } B, j \right\},$$

with value $+\infty$ if there is no admissible partition. For rational orthogonal categorical inputs this minimum and an attaining quotient are computable. If $\varepsilon_c = 0$ and $\varepsilon = 0$, the coarsest admissible partition is the future-output equivalence

$$(s, c) \sim (s', c') \iff F_{s,j}(dc) = F_{s',j}(dc') \quad \text{for every } d \in C, j,$$

where componentwise constant maps are identified with their values.

Proof. An equivariant partition induces well-defined command permutations on its blocks. Choosing a minimizing center for each $A_{B,j}$ gives exactly the stated error. Conversely, the fibers of π in any component-quotient realization form an equivariant partition. Its decoder must approximate every executable product in each corresponding component. Such products are dense in that component, so continuity extends the error bound to $A_{B,j}$. Its decoder is therefore a permitted center. This proves (6.2). The command generators suffice to check equivariance, since their images generate the finite group C .

There are finitely many partitions of X . For polynomial categorical data their radius tests are of the form (6.1), with a finite disjunction over the relevant seed/component pairs, so the minimum is computable. In the exact componentwise constant case, equality of all future outputs is an equivariant equivalence relation. Every admissible partition refines it, and its own blocks have singleton output images for each query. It is therefore the unique coarsest admissible partition. $\square$

The minimization in (6.2) is not presented as the unrestricted stationary stochastic minimum, still less as the unrestricted nonuniform minimum at the boundary. Randomized encodings can identify numerical behaviors without identifying component pairs. What is classified here is exactly the finite deterministic quotient construction that attains the radius threshold. Theorem 6.1(4) separately computes the unrestricted minimum at each fixed horizon.

7. RELATION TO NUMERICAL AUTOMATA AND POSITIVE REALIZATION

The input/output object and the order of quantifiers are decisive in comparing realization theorems. Classical probabilistic automata assign numerical acceptance probabilities to words, while stochastic-language results also study threshold recognition [13, 12]. A numerical equivalence of two fixed machines is not the assertion that for every horizon there exists a new bounded-width approximate machine. Conversely, language equivalence need not preserve the numerical probabilities.

For weighted automata over a field, the finite-Hankel-rank characterization gives finite linear representations and identifies their minimum dimension [3]. It does not enforce nonnegative rows, stochastic normalization, or simplex-valued decoders. Our common-row contraction concerns these constraints simultaneously. The invariant observable quotient in Theorem 3.4 uses the same reachable/invisible linear reduction; its additional conclusion is a finite-group criterion for exact positive realization with the full horizon-dependent quantifier.

The classical positive-realization problem asks for a positive system representing specified input/output data, and distinguishes positive dimension from unrestricted linear dimension [2]. Invariant cones and polyhedral sections are part of that theory, not new notions introduced here. Proposition 4.5 is an explicit common invariant-enclosure construction. The new matched statement pairs it with a full-subcritical lower estimate against all clocked stochastic machines. Neither one cutwise nonnegative factorization nor a finite-rank Hankel matrix supplies that lower estimate.

Brodsky and Pippenger identify bounded-error measure-once quantum languages as group languages [5, Theorem 3.3]. Their probabilistic simulation result preserves a cutpoint language with a potentially changed cutpoint [5, Theorem 3.6]; it is not uniform additive preservation of every acceptance probability. Our categorical unitary corollary fixes an entire terminal measurement law in total variation and allows horizon-dependent stochastic approximants. The distinction is already visible in the d -outcome threshold $1 - 1/d$.

The closure theorem and strict-threshold decision method of Blondel, Jeandel, Koiran, and Portier [4], and the effective Zariski-closure algorithms [6, 11], are algorithmic premises of Section 6. Computing the closure is not a new claim of this article. The additional outputs are a constrained minimax noise threshold, an attaining categorical decoder, and a terminating construction of an all-machine occupation constant. Recent Lie-algebra-based closure computations [9] give an alternative representation of the algebraic closure, but their complex connected components still require real-component analysis for the present application.

Peter–Weyl approximation, Haar averaging, and the representative-function algebra are classical compact-group and almost-periodic-function machinery [8]. Uniform approximation by a finite sum of matrix coefficients gives a finite *linear* representation of an output function. It does not give positive stochastic rows or the repeated information loss quantified by (1.3). Similarly, Steinberg’s minimal compact topological automaton and enveloping-monoid results [14, Theorems 2.2 and 2.6] concern recognition of a language. His compact topological-monoid criterion [14, Proposition 2.8] uses clopen recognition, not a continuous simplex-valued response with a prescribed error norm. Our return condition does not classify arbitrary irreversible semigroups.

Distribution-based bisimulation metrics provide robust comparisons for specified probabilistic automata [7]. They do not by themselves impose a minimum over all horizon-specific stochastic realizations of an external group experiment. The finite future-output equivalence in Theorem 6.2 is deliberately a component-quotient statement, rather than a claim that approximate equivalence always has a unique minimal positive stochastic realization.

Comparison object	Error or equivalence	Machine quantifier
Weighted series [3]	Exact scalar coefficients; linear rank	One linear representation
Group-language simulation [5]	Cutpoint language; not additive probability error	One automaton
Positive realization [2]	Exact specified numerical response	One positive system
Topological recognition [14]	Clopen language acceptance	One recognizing action
Bisimulation metrics [7]	Distributional comparison of given systems	Two specified automata
Theorem 1.2	Worst-word norm error; TV for a whole categorical law	A new clocked stochastic machine at every horizon
Theorem 4.1	Binary TV, every fixed $\varepsilon < \rho/2$	Same nonuniform lower model; exact upper construction

The contribution is the joint conclusion of a sharp constrained radius, finite permutation attainment, arbitrary-width occupation below the threshold, and, for the specified Diophantine family, a matching full-subcritical growth law. The elementary radius center, the arithmetic pigeonhole argument, the closure algorithm, and the harmonic-analysis ingredients are not asserted individually new. The accompanying theorem-level audit records the closest comparisons and their changed hypotheses; it is not a claim of exhaustive priority clearance.

7.1. Additional mathematical material. The preceding article and its complete supplement remain separately buildable. They retain the scalar two-extreme proof, compatible reachable sections, exact stationarization, one-surplus geometry, stable sections, finite-horizon certificates, word distortion and arithmetic scales, two-state duality, finite-bit compilation, and the global residual hierarchy. Their source bytes and loaded mathematical labels are preserved by a manifest. None is needed to repair a missing argument in the present main theorems. This separation preserves additional conclusions without assigning a referee of the present article the task of recertifying every historical derivation.

The finite-dimensional realization results are also logically independent of the repository's analytic A/B/C/D dependency graph. No raw local limit, stopped entropy recovery, global past kernel, Mosco limit, nonlinear resolvent, filtering, or posterior-contraction premise is imported as a proved step here. The standalone descriptive title reflects the mathematical results proved in this article; the repository series name serves only as a provenance identifier.

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